

113. M. Levoy et al., "Light field microscopy," *ACM Trans. Graph.* **25**(3), 924–934 (2006).
114. R. Horstmeyer et al., "Flexible multimodal camera using a light field architecture," in *IEEE Int. Conf. on Computational Photography*, pp. 1–8, IEEE Press (2009).
115. T. A. Mitchell and T. W. Stone, "Compact snapshot multispectral imaging system," U.S. Patent No. 8027041 B1 (2011).
116. E. le Coarer et al., "PYTHEAS: a multi-channel Fabry Perot spectrometer for astronomical imaging," *Astron. Astrophys. Suppl. Ser.* **111**(2), 359–368 (1995).
117. B. E. Bayer, "Color imaging array," U.S. Patent No. 3971065 A (1976).
118. P. E. Buchsbaum and M. J. Morris, "Method for making monolithic patterned dichroic filter detector arrays for spectroscopic imaging," U.S. Patent No. 6,638,668 B2 (2003).
119. L. Miao, H. Qi, and W. Snyder, "A generic method for generating multispectral filter arrays," in *Int. Conf. on Image Processing*, Vol. 5, pp. 3343–3346, IEEE Press (2004).
120. G. A. Baone and H. Qi, "Demosaicking methods for multispectral cameras using mosaic focal plane array technology," *Proc. SPIE* **6062**, 60620A (2006).
121. J. Brauers and T. Aach, "A color filter array based multispectral camera," presented at 12. *Workshop Farbbildverarbeitung*, pp. 5–6, Ilmenau, Germany (October 2006).
122. R. Shrestha, J. Y. Hardeberg, and R. Khan, "Spatial arrangement of color filter array for multispectral image acquisition," *Proc. SPIE* **7875**, 787503 (2011).
123. S.-W. Wang et al., "Concept of a high-resolution miniature spectrometer using an integrated filter array," *Opt. Lett.* **32**(6), 632–634 (2007).
124. J. Mercier, T. Townsend, and R. Sundberg, "Utility assessment of a multispectral snapshot LWIR imager," in *2nd Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing*, pp. 1–5, IEEE, Hoboken, NJ (2010).
125. J. M. Eichenholz et al., "Real time megapixel multispectral bioimaging," *Proc. SPIE* **7568**, 75681L (2010).
126. N. Gupta, P. R. Ashe, and S. Tan, "Miniature snapshot multispectral imager," *Opt. Eng.* **50**(3), 033203 (2011).
127. H. Halpert and B. L. Musicant, "N-color (Hg,Cd)Te photodetectors," *Appl. Opt.* **11**(10), 2157–2161 (1972).
128. A. Rogalski, J. Antoszewski, and L. Faraone, "Third-generation infrared photodetector arrays," *J. Appl. Phys.* **105**(9), 091101 (2009).
129. G. C. Gerhard, "A multispectral image sensor array," *Proc. IEEE* **59**(12), 1718 (1971).
130. M. N. Abedin et al., "Multicolor focal plane array detector technology: a review," *Proc. SPIE* **5152**, 279–288 (2003).
131. X. C. Sun et al., "Multispectral pixel performance using a one-dimensional photonic crystal design," *App. Phys. Lett.* **89**(22), 223522 (2006).
132. P. Parrein et al., "Multilayer structure for a spectral imaging sensor," *Appl. Opt.* **48**(3), 653–657 (2009).
133. P. Mitra et al., "Multispectral long-wavelength quantum-well infrared photodetectors," *Appl. Phys. Lett.* **82**(19), 3185–3187 (2003).
134. A. Longoni et al., "The transverse field detector (TFD): a novel color-sensitive CMOS device," *IEEE Electron Device Lett.* **29**(12), 1306–1309 (2008).
135. K. Kishino et al., "Resonant cavity-enhanced (RCE) photodetectors," *IEEE J. Quant. Electron.* **27**(8), 2025–2034 (1991).
136. J. Wang et al., "Cavity-enhanced multispectral photodetector using phase-tuned propagation: theory and design," *Opt. Lett.* **35**(5), 742–744 (2010).
137. F. Zaraga, G. Langfelder, and A. Longoni, "Implementation of an interleaved image sensor by means of the filterless transverse field detector," *J. Electron. Imaging* **19**(3), 033013 (2010).
138. B. Lyot, "Un monochromateur à grand champ utilisant les interférences en lumière polarisée [Wide field monochromator using polarized light interference]," *Comptes Rendus de l'Académie des Sciences* **197**, 1593–1595 (1933).
139. A. Title and W. Rosenberg, "Research on spectroscopic imaging, vol. 2: reference literature," NASA Tech. Report NSAA-CR-158702, Lockheed Palo Alto Research Laboratories, California (1979).
140. B. Lyot, "Le filtre monochromatique polarisant et ses applications en physique solaire," *Ann. d'Astrophysique* **7**, 31–79 (1944).
141. J. W. Evans, "The birefringent filter," *J. Opt. Soc. Am.* **39**(3), 229–242 (1949).
142. A. Gorman, D. W. Fletcher-Holmes, and A. R. Harvey, "Generalization of the Lyot filter and its application to snapshot spectral imaging," *Opt. Express* **18**(6), 5602–5609 (2010).
143. G. Wong, R. Pilkington, and A. R. Harvey, "Achromatization of Wollaston polarizing beam splitters," *Opt. Lett.* **36**(8), 1332–1334 (2011).
144. A. A. Wagadarikar, M. E. Gehm, and D. J. Brady, "Performance comparison of aperture codes for multimodal, multiplex spectroscopy," *Appl. Opt.* **46**(22), 4932–4942 (2007).
145. S. B. Mende et al., "Hadamard spectroscopy with a two-dimensional detecting array," *Appl. Opt.* **32**(34), 7095–7105 (1993).
146. S. T. McCain et al., "Coded aperture Raman spectroscopy for quantitative measurements of ethanol in a tissue phantom," *Appl. Spectrosc.* **60**(6), 663–671 (2006).
147. M. E. Gehm and D. J. Brady, "High-throughput hyperspectral microscopy," *Proc. SPIE* **6090**, 609007 (2006).
148. A. A. Wagadarikar et al., "Video rate spectral imaging using a coded aperture snapshot spectral imager," *Opt. Express* **17**(8), 6368–6388 (2009).
149. V. Studer et al., "Compressive fluorescence microscopy for biological and hyperspectral imaging," *Proc. Nat. Acad. Sci. U.S.A.* **109**(26), E1679–E1687 (2012).
150. E. J. Candès and Y. Plan, "A probabilistic and RIPless theory of compressed sensing," *IEEE Trans. Inf. Theory* **57**(11), 7235–7254 (2011).
151. D. S. Kittle, D. L. Marks, and D. J. Brady, "Design and fabrication of an ultraviolet-visible coded aperture snapshot spectral imager," *Opt. Eng.* **51**(7), 071403 (2012).
152. L. Gao et al., "Depth-resolved image mapping spectrometer (IMS) with structured illumination," *Opt. Express* **19**(18), 17439–17452 (2011).
153. M. W. Kudenov and E. L. Dereniak, "Compact real-time birefringent imaging spectrometer," *Opt. Express* **20**(16), 17973–17986 (2012).
154. K. Oka and T. Kaneko, "Compact complete imaging polarimeter using birefringent wedge prisms," *Opt. Express* **11**(13), 1510–1519 (2003).
155. C. F. Cull et al., "Identification of fluorescent beads using a coded aperture snapshot spectral imager," *Appl. Opt.* **49**(10), B59–B71 (2010).
156. P. B. Fellgett, "The theory of infra-red sensitivities and its application to investigations of stellar radiation in the near infra-red," Ph.D. Thesis, University of Cambridge (1951).
157. P. Jacquinot, "The luminosity of spectrometers with prisms, gratings, or Fabry-Perot etalons," *J. Opt. Soc. Am.* **44**(10), 761–765 (1954).
158. P. Griffiths, H. J. Sloane, and R. W. Hannah, "Interferometers vs monochromators: separating the optical and digital advantages," *Appl. Spectrosc.* **31**(6), 485–495 (1977).
159. L. W. Schumann and T. S. Lomheim, "Infrared hyperspectral imaging Fourier transform and dispersive spectrometers: comparison of signal-to-noise based performance," *Proc. SPIE* **4480**, 1–14 (2002).
160. R. G. Sellar, G. D. Boreman, and L. E. Kirkland, "Comparison of signal collection abilities of different classes of imaging spectrometers," *Proc. SPIE* **4816**, 389–396 (2002).
161. R. A. Keller and T. S. Lomheim, "Imaging Fourier transform spectrometer (IFTS): parametric sensitivity analysis," *Proc. SPIE* **5806**, 267–287 (2005).
162. J. Harlander, R. J. Reynolds, and F. L. Roesler, "Spatial heterodyne spectroscopy for the exploration of diffuse interstellar emission lines at far-ultraviolet wavelengths," *Astrophys. J.* **396**(2), 730–740 (1992).
163. J. M. Harlander et al., "Shimmer: a spatial heterodyne spectrometer for remote sensing of Earth's middle atmosphere," *Appl. Opt.* **41**(7), 1343–1352 (2002).
164. S. Watchorn et al., "Sunlight fluorescence observations at 589 nm with the SHIELDS spectrometer system: a progress report," *Proc. SPIE* **7812**, 781207 (2010).
165. M. W. Kudenov et al., "Spatial heterodyne interferometry with polarization gratings," *Opt. Lett.* **37**(21), 4413–4415 (2012).
166. D. F. Barbe, "Time delay and integration image sensors," in *Solid State Imaging*, pp. 659–671, NATO Advanced Science Institutes (ASI), Leyden, Netherlands (1975).
167. H.-S. Wong, Y. L. Yao, and E. S. Schlig, "TDI charge-coupled devices: design and applications," *IBM J. Res. Devel.* **36**(1), 83–106 (1992).
168. G. Lepage, J. Bogaerts, and G. Meynans, "Time-delay-integration architectures in CMOS image sensors," *IEEE Trans. Electron Devices* **56**(11), 2524–2533 (2009).
169. X.-F. He and O. Nixon, "Time delay integration speeds up imaging," *Photon. Spectra* **46**(5), 50–53 (2012).
170. J. B. Pawley, ed., *Handbook of Biological Confocal Microscopy*, 3rd ed., Springer, New York (2006).
171. N. Hagen, L. Gao, and T. S. Tkaczyk, "Quantitative sectioning and noise analysis for structured illumination microscopy," *Opt. Express* **20**(1), 403–413 (2012).
172. J. E. Sánchez et al., "Review and implementation of the emerging CCSDS recommended standard for multispectral and hyperspectral lossless image coding," in *2011 First Int. Conf. on Data Compression, Communications and Processing*, pp. 222–228, IEEE Press (2011).



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